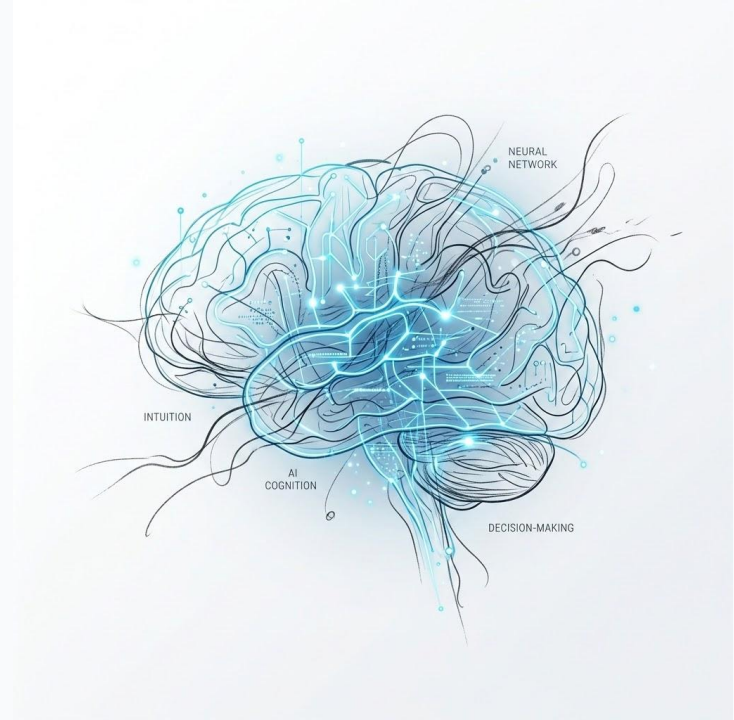


From Human Learning & Judgement to AI Systems

SYS 478. Fall 2026. Prof. Sarah Van Wart



Transitioning from Humans to “AIs”

So far, we've asked:

- How do humans learn?
- How are learning and judgment shaped by social context?
- How do values shape moral reasoning?
- Who has power to define what counts as normal, good, or correct?

Today:

- What do we need to understand about how an AI system works before we can evaluate it well?

Activity: Draw the System (20 minutes)

Draw how you think a streaming service recommends a movie based partly on what you have watched before.

As you draw, think about:

- What information does the system have access to?
- What happens to that information?
- What, if anything, changes over time?
- How does the system arrive at a recommendation?

If there is something you are unsure about, you can mark it with a “?”

Share Out

- What parts were easy to explain?
- Where did you have to guess?
- Where did you write something like:
 - “AI”
 - “Algorithm”
 - “it learns”
 - “it figures out”

What would you need to know to make your explanation more precise?

A New Question for the Course

We've been asking:

- What is learning and how does it happen (physical changes, cognitive infrastructures, social and cultural dimensions, and power dynamics)?
- How does one decide what is moral and just?

We're now going to explore this question from a computational perspective:

- What is the system actually doing?

Some caveats...

- Understanding mechanism does not settle ethical or political questions
- But it can make those questions more precise

Questions We'll Keep Asking

Dimension	Question
Representation	What information does the system use, and how is it represented?
Procedure	What steps or algorithms transform inputs into outputs?
Learning	What changes through training or experience?
Categories	What counts as similar, different, normal, or anomalous?
Values	What outcome gets rewarded or optimized?
Norms	What counts as correct or acceptable?
Power	Who chose these things, and who is affected?
Impacts	What benefits, harms, risks, or downstream effects follow?

What Can We Already Analyze?

For each question:

- Do we already have tools for answering it?
- Do we partly understand it?
- Do we need more technical, experiential, societal, or contextual knowledge?
- Possible categories:
 - Already
 - Partly
 - Not yet

Case: AI Hiring

A company uses an AI system to rank applicants for interviews

What We're Learning Next...

- **Representation (Tuesday)** → How does the world become data?
- **Procedure (Thursday)** → How does a computer actually work?
- **Learning** → How can a system change based on examples?

Understanding how things work (mechanism) will help you make your analysis more precise and powerful:

“AI hiring systems can reproduce bias.”

“If a model is trained to predict historical hiring decisions, those decisions become part of the target the model learns to reproduce.”

More Precision in Our Analysis

Understanding how things work (mechanism) will help you make your analysis more precise and powerful! Compare:

- “AI hiring systems can reproduce bias.”
- “When a model learns from past hiring decisions, it doesn't just learn from them - it learns to repeat them.”

What's one question about how AI works that you want to actually understand?

As models continue to get better, how does the training change?

Legal precedents – where is this going to go

What aspects of how AI is being implemented now makes it hard to change later.
Why is it “inevitable”?

AGI – it can learn and become sentient

What makes something AI versus a regular algorithm

Where the energy and resource use within the broader picture?

What's one question about how AI works that you want to actually understand?

How different AI would be – particles, analog computing.

AI for what? What is the goal?

- Power
- Utopia

You have a homework assignment due